

EXPERIMENT 14: 5G THROUGHPUT VS SNR WITH MODULATION LABELS

Objective – 1:

MATLAB Code:

```
clc;

clear;

close all;

% SNR values (dB)
SNR = 0:5:30;

% Maximum Throughput (Mbps)
Max_TP = 100;

% Throughput calculation

Throughput = Max_TP*(1-exp(-SNR/10));

disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(SNR,Throughput,'-o','LineWidth',2)

title('5G Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

bar(Throughput)

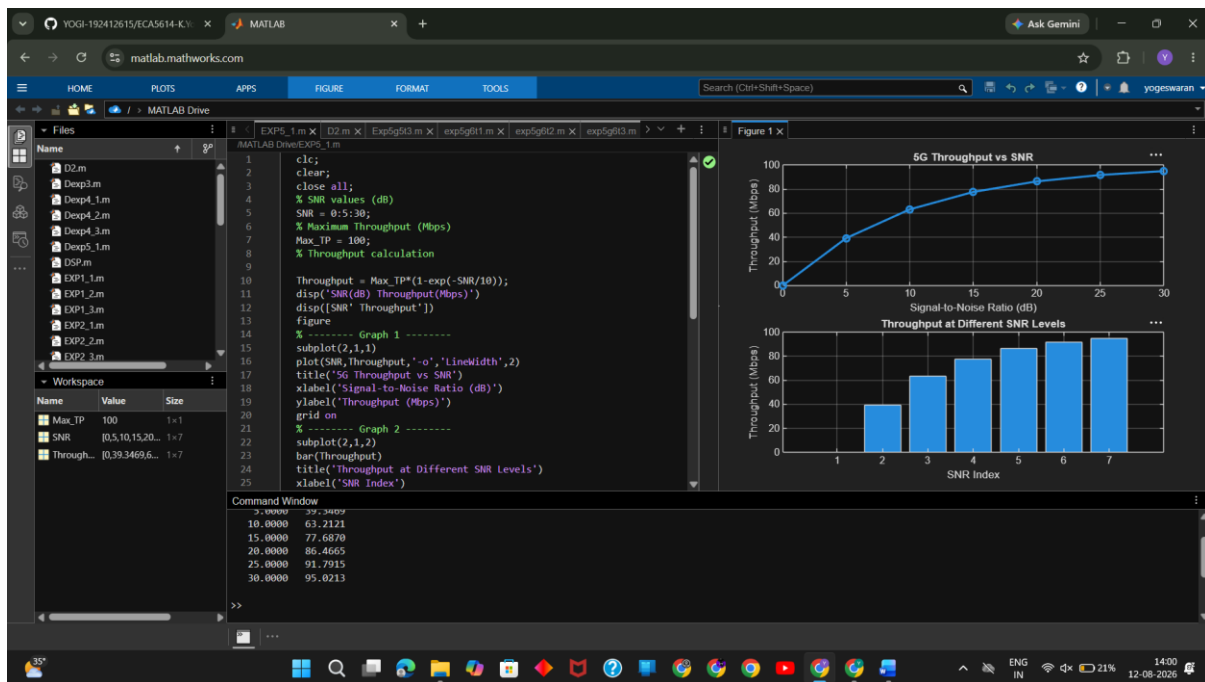
title('Throughput at Different SNR Levels')

xlabel('SNR Index')

ylabel('Throughput (Mbps)')

grid on
```

OUTPUT



Objective – 2: Compare Slice Utilization (%)

Matlab Code :

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Bit Error Rate (BER)
```

```
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
```

```
% Reliability (%)
```

```
Reliability = (1-BER)*100;
```

```
disp('SNR(dB) BER Reliability(%)\')
```

```
disp([SNR' BER' Reliability'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
semilogy(SNR,BER,'-o','LineWidth',2)
```

```
title('SNR vs Bit Error Rate in 5G')
```

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Bit Error Rate (BER)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(SNR,Reliability,'-s','LineWidth',2)

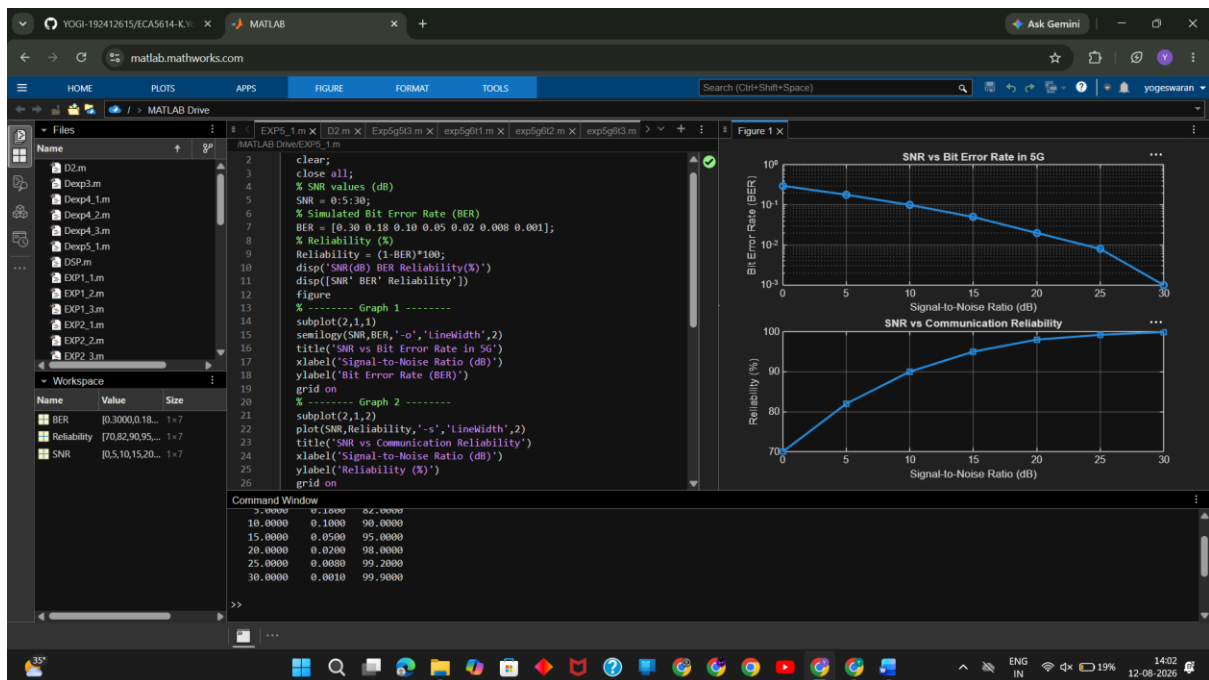
title('SNR vs Communication Reliability')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Reliability (%)')

grid on

OUTPUT



Objective – 3:

MATLAB Code:

```
clc;

clear;

close all;

% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

% Required SNR (dB)
SNR = [5 10 18 25];

% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];

disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'}))

figure

% ----- Graph 1 -----

subplot(2,1,2)

plot(SNR,Efficiency,'-o','LineWidth',2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on

% ----- Graph 2 -----

subplot(2,1,1)

plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on
```

OUTPUT

